

RUAN YIKANG

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EDUCATION

Shanghai Jiao Tong University (SJTU), Shanghai, China 2026 – 2029

M.S. in Computer Science (CS), School of Computer Science

Shanghai Jiao Tong University (SJTU), Shanghai, China 2022 – 2026

B.Eng. in Information Security (IS), School of Computer Science

B.Eng. in IS and *M.S.* student in CS, both advised by [Prof. Sun Shixuan](#).

RESEARCH INTERESTS

My research interests lie in **high-performance computing**. During my undergraduate studies, I worked on **graph computing**, particularly its integration with **serverless computing**. Currently, I am interested in high-performance LLMsys, especially in **speculative decoding** and **high-performance kernel design**.

PUBLICATION

FaaSBoard: Efficient Graph Processing with a Disaggregated Architecture on Serverless Services. Yushi Liu*, **Yikang Ruan***, Letian Ruan, Zijun Li, Sen Gao, Weihao Cui, Shixuan Sun, Quan Chen, Shuo Quan, Jie Wu, Bingsheng He, Minyi Guo. SIGMOD 2026. [Paper](#) / [Code](#)

We introduce FaaSBoard, a graph processing system with a disaggregated serverless architecture, which decouples data loading, computation, and communication, for better computing performance and cost efficiency. We develop FaaSBoard fully on AWS, using AWS Lambda, AWS Fargate, and other cloud infrastructure.

ONGOING RESEARCH FOR MEGAKERNEL

MegaKernel Reassessing Megakernel Headroom after CUDA Graphs. [Blog](#)

We characterize kernel-launch overhead in LLM serving and show that CUDA Graphs reduce host launches to 17–18 per decode step and dense-model GPU bubbles to 1–3%, indicating that Megakernel benefits do not stem simply from replacing many kernels with one.

Reconstructing Persistence with PDL.

We introduce a PDL-based approach that reconstructs Megakernel scheduling with an 81-kernel CUDA Graph. Implemented based on ThunderKittens for Llama-3.2-1B decoding on a single H100, it achieves 97.4% of the persistent Megakernel’s end-to-end throughput.

HONORS AND AWARDS

Outstanding Bachelor’s Thesis Award of the School of Computer Science (top 3%) 2026

Meng Man Wai International Exchange Fund. 2026

SJTU Undergraduate C Scholarship. 2024

Gold Award of the 2024 China International College Students Innovation Competition Shanghai Division. 2024